in complex environments. The arrival of 5G networks, and the development of future communication infrastructures such as 6G, low-earth-orbit (LEO) satellite networks, and edge computing platforms, have introduced the possibility of continuous, high-speed, and low-latency connectivity for UAV systems. These networks enable drones to perform data-intensive, long-range, and cooperative missions, making reliable communication a foundational requirement for the next phase of drone integration in civilian, industrial, and defense domains.

5G connectivity supports the transmission of high-resolution video, telemetry, environmental sensor outputs, and mission planning data in real time, allowing drones to respond instantly to changing conditions. This is especially important for beyond visual line of sight (BVLOS) operations, where drones must fly autonomously over long distances while maintaining stable command and control links. Enhanced network capacity also allows multiple drones to operate collaboratively, enabling swarm coordination, synchronized inspection, and shared mapping operations. As communication systems move toward hybrid architectures that combine terrestrial 5G networks with satellite relays and edge-cloud computing nodes, drones will gain the ability to operate seamlessly in both dense urban environments and remote terrain.

Future high-speed communication infrastructures will not only improve drone performance but also shape urban air mobility systems, where passenger drones and delivery networks rely on precise airspace coordination and continuous control. The shift to low-latency, high-capacity networks positions communication technology as the central nervous system of intelligent aerial mobility ecosystems. Thus, the advancement of drones is now inseparable from the development of next-generation